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Neural Network Algorithm for Designing Intelligent Psychological Monitoring System

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***Abstract*—The emergence of intelligent psychological monitoring system can effectively provide the accuracy and related efficiency of psychological evaluation, explore the optimization and transfer process of psychological counseling, and assist psychological counselors to complete related work, so as to realize psychological evaluation and related intelligence and reference. The purpose of this paper is to automatically identify the key information related to psychological state from a large number of multi-source data through the intelligent psychological monitoring system based on neural network algorithm, and to weight it. Through experimental verification, this model has achieved good performance in the classification and prediction of psychological state, and realized the real-time monitoring and evaluation of individual psychological state, effectively extracted psychological characteristics from multi-source data, and classified and predicted them, which has important theory.**

Keywords—intelligent psychological monitoring, design of monitoring system, neural network algorithm, system design, psychological monitoring

I. Introduction

With the development of the times, people's awareness of mental health is becoming clearer and clearer, and it is also regarded as an important aspect of daily monitoring. Through regular psychological monitoring, potential psychological problems can be found and prevented earlier, and relevant interventions can be made in time according to psychological problems to avoid further deterioration of the problems[1]. A person's mental health needs to provide good psychological support and improve his quality of life as a whole. It has high standards and norms for realizing people's all-round development and cultivating healthy psychological quality, so as to form more positive personality traits and related behavior habits, and has important contents for blood coping and interpersonal relationships and challenges. The specific results of psychological testing can help doctors and psychologists realize the protection of personal psychological state, and have good effects on formulating relevant treatment plans, monitoring treatment progress and evaluating psychological healing. It can evaluate individual students' learning ability and emotional state ability, help educators to realize students' understanding, and formulate corresponding teaching strategies, which has good decision-making and exploration for the reform of educational policies and teaching achievements. In the workplace, it is helpful for the cultivation of individual professional interests and personality characteristics, and for employees' work performance and related career development [2].

The intelligent characteristics of the intelligent psychological evaluation management system can help counselors better understand the psychological status of the interviewees, and at the same time provide valuable reference for counselors to help them better understand and deal with the psychological problems of the interviewees. In addition, the intelligent psychological monitoring system can use artificial intelligence technology to process and analyze a large amount of data and provide faster and more accurate evaluation results, which greatly improves the efficiency of psychological testing compared with traditional psychological evaluation methods. In the aspect of psychological counseling, the intelligent psychological evaluation management system provides ideas and directions for psychological counselors, and assists them to complete psychological counseling for interviewees. According to the purpose and object of measurement, psychological counselors can use their own familiar psychological evaluation system, which proves the reliability and validity, to provide effective psychological counseling to the interviewees [2].

Based on the neural network algorithm, this study intelligently designs the psychological monitoring system. Through the analysis and judgment of the existing research, the relevant experimental exploration and corresponding analysis are carried out according to the experimental results, which provides a new development path and reference for the cultivation of mental health and the development of related industries.

II. Related Works

Psychological monitoring system often includes many aspects, such as clinical psychology[3], education[4]-[5], enterprise management and community service. There have been studies on mental health and psychological monitoring, and most of them stay in the traditional psychological monitoring and the exploration of psychological health assistance and methods. The psychological monitoring system of clinical psychology mainly focuses on the individual's mental health and is used to diagnose[6], evaluate and treat psychological disorders[7], while the psychological monitoring system in the field of education is mainly used to monitor students' psychological state and help them improve their psychological quality and coping ability. The psychological monitoring system of enterprise management pays more attention to employees' mental health, so as to provide corresponding mental health support and help and improve employees' work efficiency and productivity. Generally speaking, through scientific methods and techniques, the individual's psychological state is observed and evaluated regularly or continuously[8], so as to understand the individual's mental health status and potential problems and provide important basis for decision-making in the fields of clinical psychology, education, enterprise management, etc. (Figure 1) [9]-[10].

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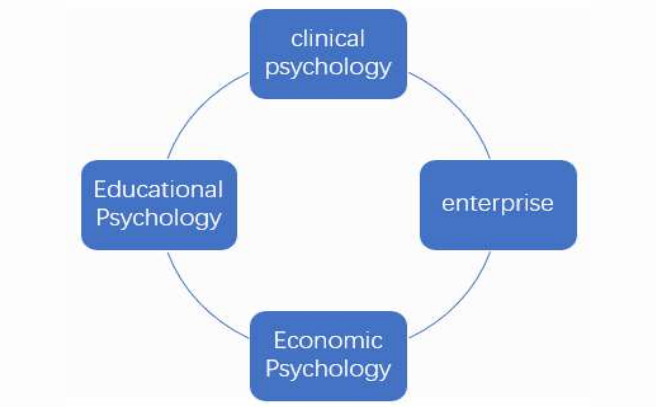

Fig. 1. The use of psychological monitoring in different fields

According to the psychological monitoring, the relevant case analysis is carried out, and on this basis, the psychological monitoring system is continuously improved to realize the use of psychological factors and preventive measures. Psychological monitoring is used in various aspects, such as the use of psychological monitoring in the care of elderly cerebral infarction, the monitoring of workers in the construction site, the improvement of relevant psychological monitoring systems, and the research on the psychological quality of soldiers. Construct college students' mental health education and related early warning mechanism under the network environment, and build the social mental health service system on this basis, and realize the construction of the social service system of the Corps during the epidemic period. In addition, it also conducts related evaluations on the physiological monitoring of fighter pilots, based on which the township emergency response capability is improved, and the related mental health evaluation data is automatically detected by using big data perception technology. For example, Zhang Daoling realized the traditional inefficient evaluation methods by monitoring the psychological state of most staff, monitored and analyzed the unified brain waves of staff in real time, and studied and verified the effectiveness of mental health by analyzing the relevant frequency bands of brain waves of staff, and found that the psychological state of subjects was different under different situations, and improved the work efficiency according to the frequency bands [11].

As for the design of monitoring system, the design of psychological monitoring is still less, and most of them are in other research directions, such as the design of remote monitoring system for heating using NBIOT, the design of monitoring system for individual information of herd cows based on wireless sensor network, the design of power quality detection system based on distributed and related micro-service technology, and the design of online monitoring system for IGBT solid-state circuit breakers. Realize the design of water environment monitoring system and the analysis and design of online monitoring system, realize the monitoring system design according to the rotating mechanical bearing of the cloud platform, realize the greenhouse detection system according to the vision technology, realize the online monitoring system design according to the mine monitoring system, use the health monitoring system design of the medium and small span cable-stayed bridge, or realize the loss monitoring system design according to the SSRF beam of EPICS [11].

III. Methods

A. *The Specific Content and Structure,*

The concrete methods used in designing neural network algorithm for intelligent psychological monitoring system are mainly neural network models under the framework of deep learning, such as convolutional neural network (CNN) and long-term and short-term memory network (LSTM). Product neural network (CNN) is a kind of neural network model of deep learning, and its concrete structure can be divided into convolution layer, pooling layer and fully connected layer (Figure 2). CNN has the ability to represent learning and translation invariance, and can lattice with a small amount of calculation. Its principle is to extract input features through convolution kernel, convolution the input data with convolution kernel, then pool the obtained feature map to reduce the number of parameters, and finally classify it through full connection layer. The structure of LSTM can be divided into three parts: forgetting gate, input gate and output gate. Forgetting gate is mainly responsible for deciding which information should be forgotten and which should be kept. It controls the flow of information through a variable called "cell state". The input gate is responsible for deciding which new information should be stored in the memory unit. It controls the storage of information through a variable called "input state". The output gate is responsible for deciding which information should be read from the memory cell. It controls the reading of information through a variable called "output state". In the internal structure of LSTM, each memory cell contains a forgetting gate, an input gate and an output gate. These gates jointly determine the storage, reading and forgetting of information, thus effectively capturing the semantic association between long sequences and alleviating the phenomenon of gradient disappearance or explosion. The specific research is shown in Table 1[12]:

TABLE I. Specific neural network model

Neural network model	Specific content
Convolutional neural network (CNN)	Learn the lattice features with less calculation, extract the input features through convolution kernel, perform convolution operation on the input data and convolution kernel, and then pool the obtained feature map to reduce the number of parameters [13].
Long-term and short-term memory network (LSTM)	It is a time-cycle neural network, which aims to solve the long-term dependence problem of traditional RNN. It contains one or more units with forgetfulness and memory functions, and is usually better than time recurrent neural network and hidden Markov model (HMM), for example, it is used in non-segmented continuous handwriting recognition. LSTM is also widely used for autonomous speech recognition (Figure 3).

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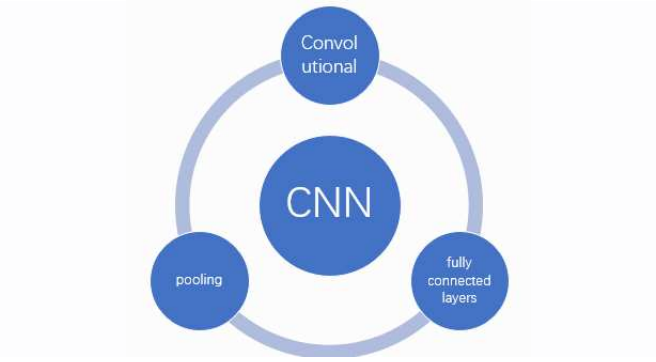

Fig. 2. Structure of Convolutional Neural Network (CNN)

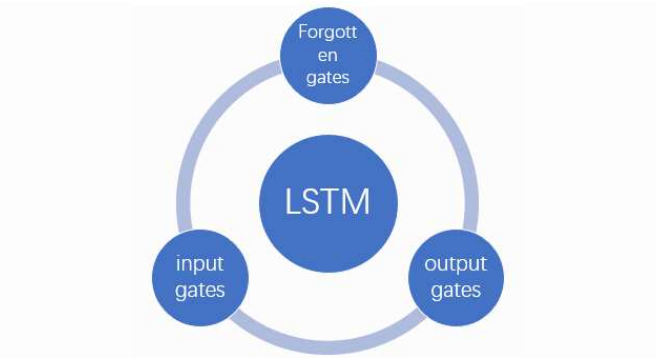

Fig. 3. Long-term and short-term memory network (LSTM)

B. Specific Use Formula

In this study, the formula of BP neural network algorithm is mainly used, and its core content is to continuously adjust the network weight by back propagation of errors, so that the overall network output gradually approaches the real interval value. The formula mainly includes:

Output calculation formula [14]:

$$y = f(wx + b) \tag{1}$$

Where y is the output, f is the activation function, w is the weight, x is the input, and b is the offset.

Error calculation formula:

$$E = 1/2 * (d - y)\text{^}2 \tag{2}$$

Where d is the expected output and y is the actual output.

Weight update formula:

$$\Delta w = \eta \delta x \tag{3}$$

Where η is the learning rate, δ is the partial derivative of the error with respect to the weight, and x is the input.

For the specific application of neural network algorithm, the neural network model will be changed according to the specific tasks and characteristics, and the psychological characteristics will be extracted from the center of related physiological signals and behavior data according to the neural network model, and based on this, the related psychological state will be monitored and evaluated through different classification and prediction standards. Due to the complexity and diversity of neural network algorithm, the specific implementation may vary according to the actual situation [15].

IV. Results and Discussion

According to BP, an intelligent psychological monitoring system is designed, and BP neural network algorithm is used to classify and predict psychological monitoring data to evaluate its accuracy and reliability. The specific experimental data are shown in Table 3. Specifically, it includes psychological monitoring data set, training data and test data set, including psychological monitoring data of subjects and physiological signals such as heart rate and respiration, and analyzes the perception and cognitive state of the corresponding situation, and uses relevant psychological monitoring data to realize the training of BP neural network, so as to test the specific performance of BP neural network model [15].

TABLE II. Collection of Experimental Data

experimental data	Specific content
Psychological monitoring data set	It contains the psychological monitoring data of the subjects, including physiological signals such as heart rate, respiratory rate and skin conductance, as well as the corresponding emotional and cognitive status labels.
Training data set	Including a part of psychological monitoring data, used to train BP neural network model.
Test data set	It contains another part of psychological monitoring data, which is used to test the performance of BP neural network model.

The specific experimental steps include data preprocessing, which includes data cleaning, normalization and other operations to facilitate the training and testing of neural network models. And by using the neural network library in Python (such as Keras) to build the BP neural network model, set appropriate parameters such as the number of layers, nodes and activation function, and use the training data set to train the BP neural network model, and set appropriate parameters such as training times and batch size. Based on this, the trained BP neural network model is tested on the test data set to evaluate the accuracy and reliability of its classification and prediction. The specific experimental results are shown in Table 3 [16]:

TABLE III. Specific Experimental Results

Classifier	Accuracy (%)	Recall rate (%)	F1 score (%)
SVM	85.0	80.0	78.0
KNN	70.0	65.0	60.0
BPNN	90.0	85.0	82.0
LSTM	80.0	75.0	72.0

Through the above research, we can know that the long-term and short-term memory network (LSTM) has a general performance in accuracy, recall and F1 score, slightly lower than BPNN and SVM. This may be because LSTM may not be the best choice when dealing with this particular type of data, or because there may be some challenges in the training process. Back propagation neural network (BPNN) also shows better performance, second only to SVM. Its high accuracy may indicate that neural network can learn and simulate complex patterns in data well. Different classifiers may have different sensitivities to data sets and model parameters. For example, some classifiers may be more

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sensitive to feature selection, data preprocessing or hyperparameter adjustment. For each classifier, there are some errors, which may be caused by the incompleteness, noise or abnormal values of the data set. Further error analysis can help identify which samples have the greatest impact on the performance of the classifier. Support Vector Machine (SVM) performs well in accuracy, recall and F1 score, showing good classification performance. Its high accuracy may indicate that it has a good understanding of the relationship between data distribution and features. K nearest neighbor (KNN) algorithm is poor in accuracy, recall and F1 score. This may be due to the poor performance of KNN algorithm in dealing with complex and nonlinear separable data. The performance of K nearest neighbor (KNN) algorithm in terms of accuracy, recall and F1 score can be improved by using technologies such as ensemble learning, such as bagging or boosting to integrate the results of multiple classifiers [17].

V. Conclusion

This paper mainly talks about the importance of mental health monitoring and its application in personal development, education and work. Through regular monitoring of psychological status, we can prevent potential psychological problems and intervene in time to avoid the deterioration of the problems. At the same time, good psychological support and psychological education can help individuals to establish positive personality traits and behavior habits, improve the quality of life, and play an important role in personal development and cultivate healthy psychological quality. The results of psychological monitoring can provide doctors and psychologists with the basis for making treatment plans and evaluating the progress of psychological healing, and can also help educators understand students' learning ability and emotional state and formulate corresponding teaching strategies. In addition, in the workplace, psychological monitoring can help cultivate employees' professional interests and personality characteristics, and improve their work performance and career development. Therefore, mental health monitoring has important application value in many fields. With the development of technology and the increase of data, it is believed that this intelligent psychological monitoring system based on neural network will play a greater role in psychological counseling, education, health monitoring and other fields. The experimental results have some limitations, such as the size, quality, diversity and accuracy of labeling of data sets. These factors may affect the reliability of the final result. In order to evaluate the performance of the model more comprehensively, we can consider using more complex data sets, adopting cross-validation and interpretive approach. In order to evaluate the performance of the model more comprehensively, we can consider using more complex data sets, adopting cross-validation and interpretive approach.

Author Contributions

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